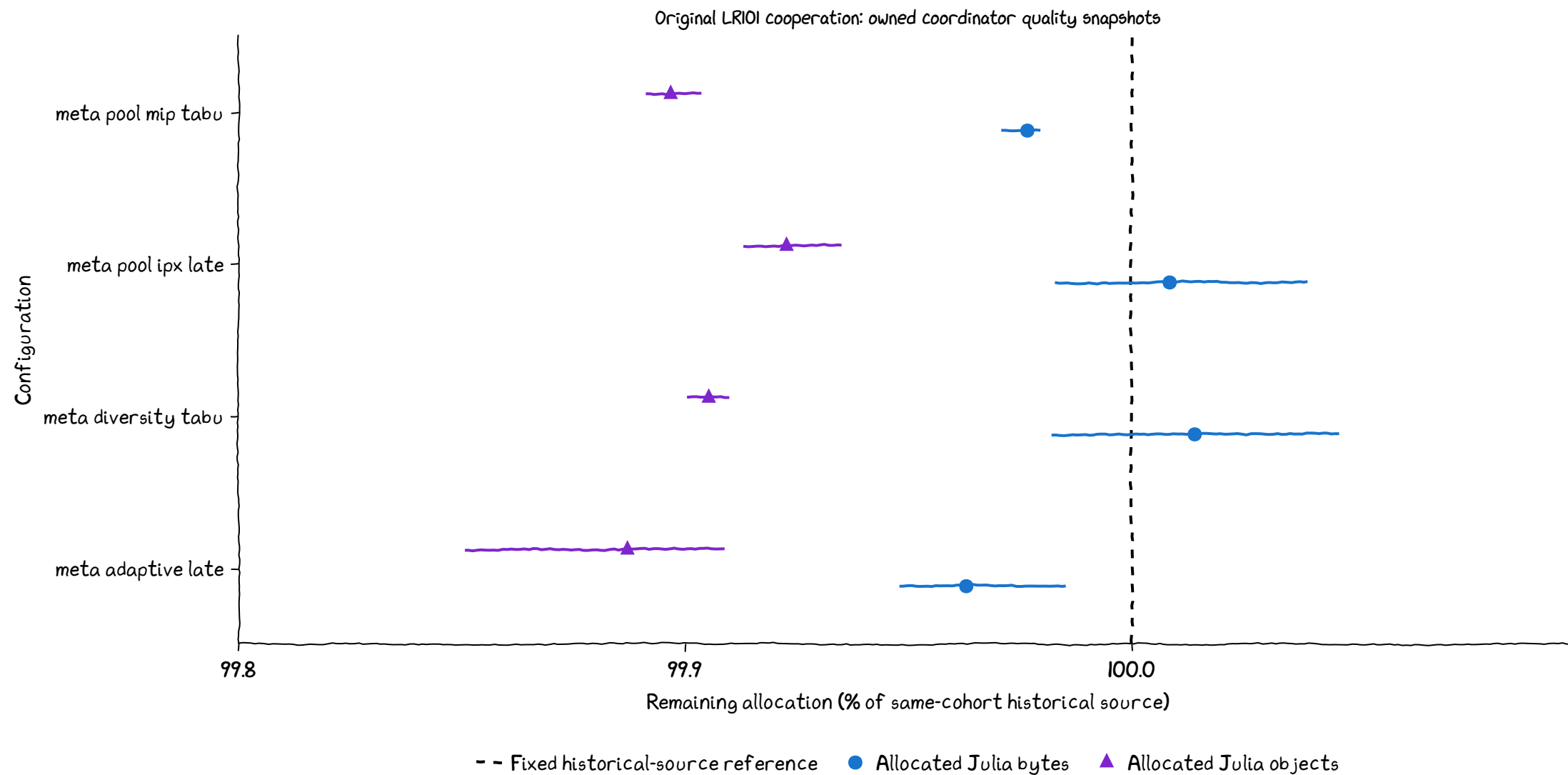




12 paired paths: 4 configurations × 3 seeds; 8 actual cooperative episodes × 32 steps × 2 workers. Four small byte increases retained; all object counts decrease. Markers: mean paired ratio; bars: full seed range. Complete substantive trace/routes/RNG work and original validators match; zero compilation/recompilation. Frozen V7 integration remains provisional for a separate generic QUBO constructor regression. Concurrent timing; no controlled speed or quality claim.